Sherman Lim

PhD student, Carnegie Mellon University
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RESEARCH INTERESTS

Memory systems, novel memory technologies, LLM inference systems, hardware-software co-design.

EDUCATION

Carnegie Mellon University — Pittsburgh, PA
PhD in Computer Science — Aug 2025 – Present

- Advisor: Prof. George Amvrosiadis
- Expected graduation: May 2030

National University of Singapore — Singapore
Bachelor of Computing (Computer Science) with Highest Distinction — Aug 2019 – Jan 2023

- GPA: 4.88 out of 5.00
- Thesis: Congestion Control Speciation in QUIC
- Advisor: Prof. Ben Leong

PUBLICATIONS

Sidharth Sankhe, Felix Zhang, Umayrah Chonee, **Sherman Lim**, Jiasheng Hu, Jialin Li, and Qizhen Zhang. *"PD3: Prefetching Data with DPUs for Disaggregated Memory"*. In Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26). May 2026.

Ayush Mishra, **Sherman Lim**, and Ben Leong. *"Understanding Speciation in QUIC Congestion Control"*. In Proceedings of the 22nd ACM Internet Measurement Conference (IMC '22). October 2022.

RESEARCH EXPERIENCE

Carnegie Mellon University — Pittsburgh, PA
Graduate Research Assistant, Parallel Data Lab — Aug 2025 – Present
Advisor: Prof. George Amvrosiadis

- Conceived and leading a project co-designing LLM inference systems for accelerators with emerging High Bandwidth Flash (HBF) memory.
- Developing a prototype on vLLM with emulated HBF-integrated GPUs.

National University of Singapore — Singapore
Research Assistant — Apr 2025 – Jul 2025
Advisors: Prof. Jialin Li and Prof. Qizhen Zhang

- Designed a high-throughput component on resource-constrained NVIDIA BlueField DPUs that detects disaggregated-memory cache misses to enable prefetching in PD3 (NSDI '26).

National University of Singapore — Singapore
Research Assistant — Jan 2021 – Nov 2022
Advisor: Prof. Ben Leong

- Built QUICbench, a tool for benchmarking congestion control implementations across QUIC stacks, and used it to show that deployed QUIC CCAs diverge greatly from their TCP namesakes (IMC '22).

INDUSTRY EXPERIENCE

Jump Trading — Singapore
Software Engineer — Jan 2023 – Feb 2025

- Led development of latency-critical C++ systems for high-frequency trading spanning market risk management, market data ingestion, order submission, FPGA interfacing, and embedded environments.
- Co-designed a low-latency inter-process communication library tailored to a specific CPU microarchitecture.
- Mentored a summer intern, scoping and supervising a performance benchmarking project.

Jump Trading Singapore
Software Engineer Intern May 2022 – Jul 2022

- Designed and implemented a C++ framework for live-testing order submissions.

ByteDance Singapore
Software Engineer Intern May 2021 – Jul 2021

- Developed a Java database driver library for a data warehouse.

SKILLS

C++, C, CUDA, Python, Java, x86-64 assembly, vLLM, NVIDIA BlueField DPUs, Linux

TEACHING ASSISTANTSHIP

CS3210 Parallel Computing, NUS Fall 2022

CS2106 Introduction to Operating Systems, NUS Spring 2022

CS2040S Data Structures & Algorithms, NUS Spring 2021

CS2040S Data Structures & Algorithms, NUS Fall 2020

AWARDS

NUS Merit Scholarship, NUS 2019 – 2023
Full-ride, merit-based scholarship for undergraduate study at NUS.

Outstanding Undergraduate Researcher Prize (Individual), NUS 2022
Annual award that recognizes top undergraduate researchers in NUS.